

INTELLIGENT SYSTEMS & ROBOTICS

ASSIGNMENT 1

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Objective

Understand the workings of PyTorch and be able to get a good sense of the differences and similarities between TensorFlow and PyTorch

Submissions

- One submission per team
- Jupyter Notebook &
- HTML/PDF knitted version of the Notebook

Assignment (Leaf Classification)

We will use the assignment from PML (Leaf Classification) and replicate the same using PyTorch. Please make sure to follow best practices and detail your assumptions and findings within the notebook.

- Dataset: <https://www.kaggle.com/c/leaf-classification>
- Conduct your analysis using a cross-validation design.
- Conduct EDA.
- Build at least three CNN models based on hyperparameter tuning.
- Evaluate goodness of fit metrics.
- Build multi-class ROC and Precision / Recall graphs.
- Once you have your best-performing models, classify the test data
- Discuss your model's performance.

References

- https://pytorch.org/tutorials/beginner/deep_learning_60min_blitz.html
- <https://cs230.stanford.edu/blog/pytorch/>
- <https://github.com/amanchadha/stanford-cs231n-assignments-2020/blob/master/assignment2/PyTorch.ipynb>
- <https://github.com/ritchieng/the-incredible-pytorch>